PHASE

13

File Systems: VFS, initrd and TurkFS

Turn a pile of numbered blocks into named files and directories, behind one interface that also covers devices and a read-only boot image

Overall progress toward Turk-OS 1.0 (this phase in red) 83 % → 90 %

Duration	3–4 weeks
Difficulty	★★★★★ (5/5)
You need	Phase 12: block devices and the buffer cache; Phase 11: processes, `fork`/`exec`; Phase 9: mutexes
You will have	A virtual file system layer (in-memory inodes, open-file objects, per-process descriptor tables, path lookup, mount points), a read-only root file system from a tar archive, `/dev` with `tty`, `null`, `zero` and disks, and TurkFS: a writable on-disk file system in the MINIX V3 format that Linux's `fsck.minix` accepts as clean. System calls: `open`, `close`, `read`, `write`, `lseek`, `dup`, `dup2`, `stat`, `fstat`, `getdents`, `mkdir`, `unlink`, `chdir`, `getcwd`.

1 Why this phase

A disk is a very long array of blocks. Users want named files in a tree of directories that they can open, read, grow and delete, and that survive crashes. The **file system** bridges the two, and it is the largest subsystem you will write. It uses nearly everything from before: the buffer cache and block devices (Phase 12), processes and `fork` semantics (Phase 11), user-pointer checks (Phase 10), and mutexes, because file-system operations sleep on disk I/O while holding locks (Phase 9).

The phase has two halves. The **VFS** (virtual file system) is the file-system-independent layer: file descriptors, open files, path names, mount points. It is what makes "everything is a file" possible in UNIX, since a terminal, a disk and a regular file all look the same to `read`. Below it you plug in concrete file systems: first a trivial read-only one for booting, then **TurkFS**, a real on-disk format with inodes, bitmaps and indirect blocks. TurkFS uses the MINIX V3 layout from OSDI, so the book's figures describe your disk exactly, and Linux can check your images and even mount them.

2 Concepts you need

2.1 Files, directories and descriptors

OSTEP chapter 39 describes the interface. A **file** is an array of bytes with a low-level name, its **inode number**. A **directory** is a file whose contents are a list of (name, inode number) pairs, and directories nest into a tree. A process doesn't use inodes directly; `open` returns a small integer, the **file descriptor**, which indexes the process's descriptor table. Three levels of objects are involved (Figure 1), and keeping them apart is the key to getting `fork` and `dup` right. The descriptor table is per process. An **open file** records the current offset and access mode and can be shared, so after `fork` the parent and child share one offset. An **in-memory inode** exists once per file no matter how many times it is open.

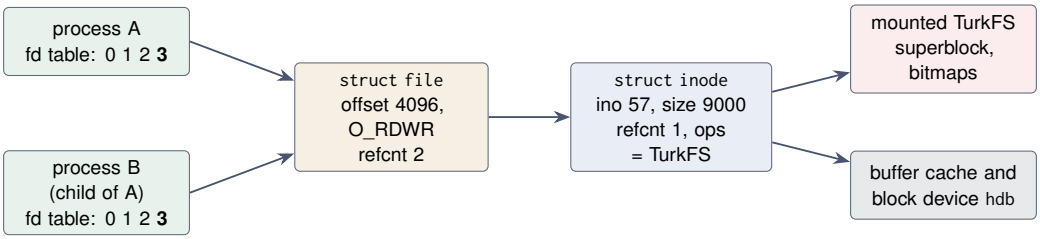


Figure 1: After `fork`, descriptor 3 in both processes refers to the same open file, and so to the same offset.

2.2 How an on-disk file system is laid out

OSTEP chapter 40 builds a very simple file system, *vsfs*, and TurkFS is its real-world sibling. The disk is divided into regions: a **superblock** describing the whole file system, an **inode bitmap** and a **zone bitmap** marking which inodes and data blocks are in use, the **inode table**, and the **data** region (Figure 2). MINIX calls data blocks *zones*; with the settings you will use, a zone is exactly one 1 KiB block (OSDI §5.6.2).

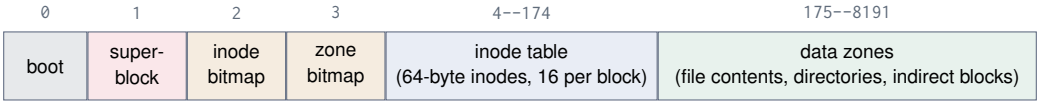


Figure 2: TurkFS on an 8 MiB image, exactly as `mkfs.minix -3` lays it out: 2736 inodes, first data zone 175. Numbers are 1 KiB block numbers.

Structure	Format (all little-endian)
superblock (at byte 1024)	`u32 s_ninodes; u16 pad; u16 s_imap_blocks; u16 s_zmap_blocks; u16 s_firstdatazone; u16 s_log_zone_size; u16 pad; u32 s_max_size; u32 s_zones; u16 s_magic (0x4D5A); u16 pad; u16 s_blocksize (1024); u8 s_disk_version`
inode (64 bytes)	`u16 i_mode; u16 i_nlinks; u16 i_uid; u16 i_gid; u32 i_size; u32 i_atime, i_mtime, i_ctime; u32 i_zone[10]`
directory entry (64 bytes)	`u32 inode; char name[60]` (name padded with zeros; inode 0 means an empty slot)
bitmaps	bit n of the inode map is inode n; bit n of the zone map is zone $n + $ `s_firstdatazone` $- 1$; bit 0 of both is reserved and always set

2.3 Finding a file's data

An inode has ten zone pointers (Figure 3). The first seven point directly at data zones, so small files need nothing else. Pointer 7 points to a **single indirect** block holding 256 more zone numbers, and pointer 8 to a **double indirect** block holding 256 pointers to blocks of 256 zone numbers each. The tenth pointer (triple indirect) is reserved and TurkFS doesn't use it. OSTEP §40.3 and OSDI §5.6.4 explain the scheme, and MOS §4.3.2 compares it with the alternatives: contiguous allocation, linked lists and FAT.

2.4 Path lookup, and what it costs

To open `/bin/hello`, the kernel starts at the root inode (inode 1), reads the root directory's data and searches it for `bin`, reads inode `bin`, reads its data and searches for `hello`, then reads inode `hello`. OSTEP §40.6 draws this as a timeline of disk reads, and it is the best way to understand why caching matters: without the buffer cache, every `open` costs several disk reads just to walk the path.

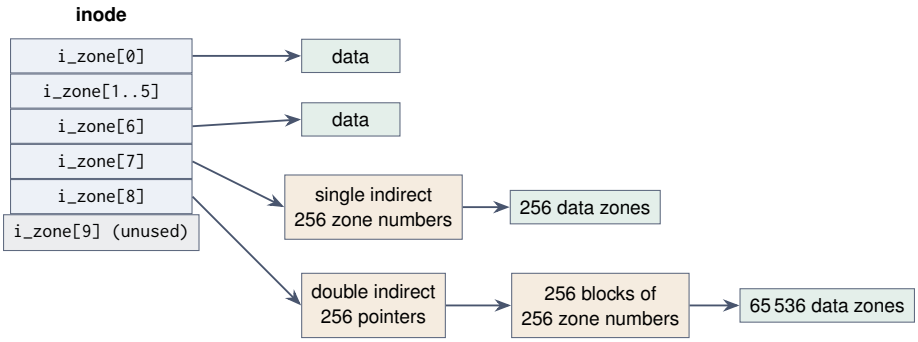


Figure 3: Zone pointers in a TurkFS inode, with 1 KiB zones and 4-byte zone numbers.

2.5 Crashes

Creating a file changes several blocks: the inode bitmap, the inode, the directory, maybe the zone bitmap and a data block. If the machine stops half-way, the file system is left inconsistent: an inode marked used that no directory points to, or a zone used by two files. OSTEP chapter 42 presents the two classic answers. **fsck** scans the whole disk after a crash and repairs it, which is simple but slow. **Journaling** writes a description of the change to a log first and replays it after a crash. TurkFS relies on `fsck.minix` on the host for now and orders its writes carefully (allocate, then initialise, then link), so a crash leaves at worst leaked blocks instead of corruption. A journal is a stretch goal.

3 Reading plan

Priority	Book	What to read	Pages	Why now
REQUIRED	LOB	Ch. 12 *File Systems*: a read-only file system, inodes, a virtual file system	69–70	The route of this phase in two pages.
REQUIRED	OSTEP	Ch. 39 *Interlude: Files and Directories*	457–476	The interface: descriptors, offsets, `fork` and `dup` sharing, links.
REQUIRED	OSTEP	Ch. 40 *File System Implementation* (vsfs, inodes, multi-level index, the access-path timeline, caching)	477–494	The design of TurkFS in miniature.
REQUIRED	OSTEP	Ch. 42 *Crash Consistency: FSCK and Journaling*	507–526	Why write order matters.
REQUIRED	OSDI	§5.6 *Overview of the MINIX 3 File System*: messages, layout, bitmaps, i-nodes, block cache, directories and paths, descriptors, locking, pipes, the READ walkthrough	548–566	This is your on-disk format, explained by its designers.
RECOMMENDED	OSDI	§5.7 *Implementation of the MINIX 3 File System*	566–606	Working code for every operation in Tasks 13.5–13.8.
RECOMMENDED	MOS	§4.1–4.3 *Files*, *Directories*, *File-System Implementation* (incl. journaling and VFS)	265–299	Allocation methods compared; VFS in §4.3.7.
RECOMMENDED	MOS	§4.5 *Example File Systems* (MS-DOS FAT, UNIX V7)	320–331	Two real designs next to yours.
RECOMMENDED	OSC	Ch. 13 *File-System Interface*; §14.1–14.5; §15.5 *Virtual File Systems*	529–582, 603–605	A thorough second treatment.

Priority	Book	What to read	Pages	Why now
OPTIONAL	OSTEP	Ch. 41 *Locality and the Fast File System*; Ch. 43 *Log-structured File Systems*	495–506, 527–542	How layout decisions affect speed.
OPTIONAL	MOS	§4.4 *File-System Management and Optimization*	299–320	Block size, free-space tracking, consistency checks, caching.
RECOMMENDED	LKD	Ch. 13, *The Virtual Filesystem*	261–288	Superblock, inode, dentry and file objects: the model for your VFS.

Real-system lens: how MINIX 3 does it

The MINIX file system server in `servers/fs/` implements the exact format you are writing. Read it alongside your code: the superblock and bitmap allocation (`alloc_bit`, `free_bit`) in `super.c (23300)`; the in-memory inode table (`get_inode`, `put_inode`, `alloc_inode`) in `inode.c (22900)`; path lookup (`eat_path`, `advance`, `search_dir`) in `path.c (26300)`; file position to zone mapping (`read_map`) in `read.c (25000)`; `open` and `mkdir` in `open.c (24500)`; descriptors in `filedes.c (23700)`. OSDI §5.6.10 traces one `read` call through all of it.

Linux, for comparison. LKD Linux's VFS has four main objects: the superblock, the inode, the **dentry** (a cached path component) and the file. The dentry cache is what makes repeated path lookups fast; your VFS has the other three.

4 Build it

Task 13.1 The VFS objects

Define the three core structures. `struct inode` is the in-memory inode, file-system independent, with a table of operations that each file system fills in. `struct file` is an open file. A mounted file system is a `struct superblock` with its root inode. Keep in-memory inodes in a table keyed by (superblock, inode number), so each file has exactly one, with a reference count.

include/vfs.h

```
struct inode_ops {
    int     (*lookup)(struct inode *dir, const char *name, struct inode **out);
    int     (*create)(struct inode *dir, const char *name, uint16_t mode, struct inode **out);
    int     (*mkdir)(struct inode *dir, const char *name, uint16_t mode);
    int     (*unlink)(struct inode *dir, const char *name);
    int     (*readdir)(struct inode *dir, uint32_t index, struct dirent *out);
    int32_t (*read)(struct inode *ino, void *buf, uint32_t n, uint32_t off);
    int32_t (*write)(struct inode *ino, const void *buf, uint32_t n, uint32_t off);
    int     (*truncate)(struct inode *ino, uint32_t size);
};

struct inode {
    uint32_t               ino;        /* number inside its file system        */
    uint16_t               mode;       /* type bits (dir, file, device) + rwx  */
    uint32_t               size;
    int                    refcnt;     /* in-memory references                 */
    struct superblock      *sb;
    const struct inode_ops *ops;
    struct mutex           lock;       /* held during operations that may sleep */
    struct inode           *mounted;   /* root of a file system mounted here   */
    void                   *priv;      /* file-system specific data            */
};

struct file {
```

```
    struct inode *inode;
    uint32_t      offset;
    int           flags;          /* O_RDONLY, O_WRONLY, O_RDWR, O_APPEND */
    int           refcnt;         /* descriptors pointing here            */
};
```

Task 13.2 Descriptors and the basic calls

Give each process a table of 32 `struct file` pointers. `open` resolves the path, allocates a `struct file` and returns the lowest free descriptor. `read` and `write` copy through a kernel buffer with `copy_to_user`/`copy_from_user` and advance the offset. `close` drops a reference. `dup` and `dup2` copy a pointer and increment the count. `fork` copies the whole table, incrementing every count; `exec` keeps it; `exit` closes everything. Keep using the Linux i386 numbers.

Call	No.	Call	No.	Call	No.
open	5	lseek	19	stat	106
close	6	mkdir	39	fstat	108
unlink	10	dup	41	getdents	141
chdir	12	dup2	63	getcwd	183

Task 13.3 Devices as files

Write a tiny `devfs` and mount it at `/dev`. Its inodes carry device operations instead of disk data: `/dev/tty` reads a line from the keyboard and writes to the console; `/dev/null` discards writes and returns end-of-file; `/dev/zero` returns zeros; `/dev/hda` and `/dev/hdb` read and write their block devices. When the kernel starts `init`, it opens `/dev/tty` three times, so `init` and everything it starts inherit descriptors 0, 1 and 2. `sys_write` to descriptor 1 now goes through the VFS like any other file.

Task 13.4 A read-only root from a tar archive

The simplest file system to parse is a tar archive in the POSIX *ustar* format, so use one as the initial root. Each file is a 512-byte header followed by its contents padded to a multiple of 512 bytes, and the archive ends with two zero blocks. The header holds the name (100 bytes at offset 0, plus a prefix of 155 bytes at offset 345), the size as an octal ASCII string (12 bytes at offset 124), the type (one byte at 156: `'0'` for a file, `'5'` for a directory) and the magic `"ustar"` at offset 257. Build the archive from a `rootfs/` directory in the Makefile, load it as a boot module, and mount it at `/`. Now `execve` can take real paths.

Makefile (root archive)

```
$(BUILD)/initrd.tar: $(USER_PROGS) $(shell find rootfs -type f)
    mkdir -p rootfs/bin && cp $(USER_PROGS) rootfs/bin/
    tar --format=ustar -cf $@ -C rootfs .
```

Verify: `execve("/bin/hello", ...)` loads the program from the archive, and `cat /etc/motd` (from the monitor for now) prints a file you put in `rootfs/etc/`.

Task 13.5 Path lookup

Write `namei(path, &inode)`. Start at the root for absolute paths and at the process's current directory otherwise, then take one component at a time: skip empty components (`//`), stay put on `.`, and call the directory's `lookup` for everything else, including `..`. When the result has a file system mounted on it, continue at that file system's root; when `..` is taken at the root of a mounted file system, go to the mount point's parent. Also write `nameiparent` (the directory and the last component), which `create`, `mkdir` and `unlink` need. Add `chdir` and `getcwd`.

Verify: `/bin/../etc/./motd`, `//etc//motd` and a relative `etc/motd` from `/` all resolve to the same inode; `/dev/..` is `/`.

Task 13.6 TurkFS, reading

Create a file-system image on the host and attach it to QEMU as the primary slave (`hdb`); your ATA driver needs to select the slave with `0xF0` in the drive register. Mount it at `/mnt`. Read the superblock through the buffer cache and check the magic and block size; read inodes by number (inode n is in block $2 + \text{imap} + \text{zmap} + (n-1)/16$, at offset $((n-1) \bmod 16) \times 64$); map a file offset to a zone through the direct, single and double indirect pointers; search directories entry by entry.

```
terminal (host)
dd if=/dev/zero of=build/turkfs.img bs=1024 count=32768     # 32 MiB
mkfs.minix -3 build/turkfs.img                               # MINIX V3 layout, 1 KiB blocks
sudo mount -o loop build/turkfs.img /mnt                     # optional: put files on it
sudo cp -r rootfs/* /mnt && sudo umount /mnt
qemu-system-i386 ... -drive file=build/turkfs.img,format=raw,if=ide,index=1,media=disk
```

Verify: `ls /mnt` (with the monitor, or a test program using `getdents`) lists what you copied from Linux, and reading a file larger than 263 KiB gives the right contents, which exercises the double indirect block.

Task 13.7 TurkFS, writing

Now allocation. Find a free inode by scanning the inode bitmap; find a free zone by scanning the zone bitmap and remember the offset: bit n is zone $n +$ `s_firstdatazone` $- 1$. Implement `create` (allocate and initialise an inode, then add a directory entry), `write` (allocate zones and indirect blocks as the file grows, zeroing new blocks), `mkdir` (a new directory containing `.` and `..`, with the parent's link count incremented), `unlink` (remove the entry, decrement the link count, and free the inode and its zones when the count reaches zero and nobody has the file open), and `truncate`. Do every change through the buffer cache, marking buffers dirty, and order the writes so a crash leaves at worst leaked space: allocate, initialise, and only then link. Add a `sync` system call and flush all file systems at shutdown.

Verify: After Turk-OS creates, writes and deletes files, shut it down cleanly and run `fsck.minix -f build/turkfs.img` on the host: it reports no errors and exits with status 0.

Task 13.8 stat, fstat and getdents

Define `struct stat` (device, inode, mode, link count, size, times) and `struct dirent` (inode number, name), and implement the three calls through the inode operations. These are what `ls` will use in Phase 14.

Task 13.9 Tests

Add these to `make test`, running on a fresh image each time; after the run, `make test` also runs `fsck.minix` on the image.

Test	What must hold
`many_files`	create 200 files with random contents, read them back, delete them; free inode and zone counts return to their starting values
`big_file`	write and verify a 2 MiB file (well into the double indirect range), then truncate it to 0
`dirs`	nested `mkdir`, removing directories, link counts of parents
`paths`	`.`, `..`, `//`, relative paths, crossing into and out of `/mnt` and `/dev`
`fd_sharing`	after `fork`, parent and child share an offset; after `dup2`, two descriptors share one; closing one leaves the other working
`readonly_root`	writing to the tar root fails with `-EROFS`

5 Testing and debugging

When the on-disk format goes wrong, compare with what Linux sees. Mount the image read-only on the host (`sudo mount -o loop,ro`) and look around, run `fsck.minix -fsv` for a detailed report, and dump raw blocks with `hexdump -C -s $((BLOCK*1024)) -n 1024 build/turkfs.img`.

Symptom	Likely cause	What to do
`fsck.minix` reports zones marked in use but not used, or the reverse	The zone bitmap offset is wrong	Bit n is zone $n + $ `s_firstdatazone` $- 1$, not zone n.
Directory listing shows garbage names or skips entries	Inode 0 slots not skipped, or names treated as NUL-terminated when they are exactly 60 bytes	Skip entries with inode 0; copy at most 60 bytes and terminate yourself.
Wrong inode read	Inode numbers start at 1	Use $n - 1$ when computing the block and offset.
Writes vanish after reboot	Buffers not marked dirty, or no `sync` at shutdown	`bdirty` after every change; flush in the shutdown path.
After `fork`, the two processes read the file independently	The descriptor table was deep-copied into new `struct file` objects	Copy pointers and increment counts; they must share the offset.
Deadlock while creating a file	A directory inode lock and a buffer lock taken in different orders in different paths	Add the file-system locks to the lock order from Phase 9: inode, then buffer.

6 Milestone

Milestone: Files and directories

- ☐ The VFS resolves paths across `/` (tar), `/dev` (devfs) and `/mnt` (TurkFS).
- ☐ Every process has descriptors 0–2 on `/dev/tty`; `fork`, `dup2` and `exec` handle descriptors correctly.
- ☐ Programs are loaded by path from the file system.
- ☐ TurkFS reads and writes files and directories, and `fsck.minix -f` on the host finds no errors.
- ☐ All file-system tests pass, including the 2 MiB file.

Self-check questions

1. List the disk reads needed to open and read `/home/turk/notes.txt` with an empty cache, following OSTEP §40.6.

2. What is the largest file TurkFS can hold with seven direct zones, one single and one double indirect block, 1 KiB zones and 4-byte zone numbers?
3. What is the difference between a file descriptor, an open file and an inode? Which of them does `fork` copy?
4. What does the VFS layer buy you? Name three things in Turk-OS that would be harder without it.
5. Why is crash consistency hard? In what order should `create` write its blocks, and why?
6. What is the difference between a hard link and a symbolic link? How does the link count relate to `unlink`?

7 Going further

Make TurkFS the root file system and keep the tar archive only for early boot. Add a simple metadata journal (OSTEP §42.3): write the blocks of each operation to a log region first, then a commit block, then apply them, and replay the log at mount. Write a FAT16 driver (MOS §4.5.1) so Turk-OS can read USB-stick images made on any computer. Or add symbolic links, `rename`, and file permissions checked against the process's user ID, which Phase 15 builds on.

Next: **Phase 14 Userland: libc, Shell and Utilities** →

Books used in this plan

LOB — Erik Helin, Adam Renberg. *The little book about OS development*, 2015. The hands-on x86 guide. Free at https://littleosbook.github.io.

OSTEP — Remzi H. Arpaci-Dusseau, Andrea C. Arpaci-Dusseau. *Operating Systems: Three Easy Pieces*, v0.90, Arpaci-Dusseau Books, 2015. Main theory text. Free at https://pages.cs.wisc.edu/~remzi/OSTEP/.

OSC — A. Silberschatz, P. B. Galvin, G. Gagne. *Operating System Concepts*, 10th ed., Wiley, 2018. Reference theory text.

MOS — Andrew S. Tanenbaum, Herbert Bos. *Modern Operating Systems*, 4th ed., Pearson, 2015. Hardware, I/O, design, security.

OSDI — Andrew S. Tanenbaum, Albert S. Woodhull. *Operating Systems: Design and Implementation*, 3rd ed., Pearson Prentice Hall, 2006. MINIX 3 with full source listing.

LKD — Robert Love. *Linux Kernel Development*, 3rd ed., Addison-Wesley, 2010. How Linux implements each subsystem.

LKP — Kaiwan N. Billimoria. *Linux Kernel Programming*, Packt, 2021. Modern kernel practice: logging, allocators, the scheduler, synchronisation.

The books do not come with this plan. LOB and OSTEP are free online; buy or borrow the other five (the roadmap lists their ISBNs). Where books disagree, the plan follows the newer one when its method is also better for Turk-OS. Page numbers are the *printed* page numbers of these editions, not PDF page indices. With another edition, or with the free OSTEP chapters (each one starts at page 1), find the reading by its chapter and section numbers. MINIX source references like `kernel/proc.c (07400)` give the file and its starting line in OSDI Appendix B.